

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 | 5 |   |   |
| A | <div>12V Power Input</div> <div></div> <div>File: POS12_power_input.kicad_sch</div> <div>12V Telemetry</div> <div></div> <div>File: POS12_telemetry.kicad_sch</div> <div>3.3V Power Supply</div> <div></div> <div>File: POS3P3_power_supply.kicad_sch</div> <div>3.3V Telemetry</div> <div></div> <div>File: POS3P3_telemetry.kicad_sch</div> <div>LED Power Supply</div> <div></div> <div>File: led_power_supply.kicad_sch</div> <div>LED Telemetry</div> <div></div> <div>File: led_telemetry.kicad_sch</div> <div>dsPIC33 Programming</div> <div></div> <div>File: dspic33_programming.kicad_sch</div> <div>dsPIC33 Power</div> <div></div> <div>File: dspic33_power.kicad_sch</div> <div>dsPIC33 GPIO</div> <div></div> <div>File: dspic33_gpio.kicad_sch</div> <div>USB UART Bridge</div> <div></div> <div>File: usb_uart_bridge.kicad_sch</div> |   | <div>Platform ETC</div> <div></div> <div>File: platform_etc.kicad_sch</div> <div>PGOOD LEDs</div> <div></div> <div>File: pgood_leds.kicad_sch</div> <div>Status LEDs</div> <div></div> <div>File: status_leds.kicad_sch</div> <div>RCA Signal Conditioning</div> <div></div> <div>File: rca_signal_conditioning.kicad_sch</div> <div>Speaker Signal Conditioning</div> <div></div> <div>File: speaker_signal_conditoning.kicad_sch</div> <div>Filtering, PGAs</div> <div></div> <div>File: filtering_pgas.kicad_sch</div> <div>Power Button</div> <div></div> <div>File: power_button.kicad_sch</div> |   |   |   | A |
| B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   | B |   |
| C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   | C |   |
| D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   | D |   |
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 | 5 |   |   |

Sheet: /  
File: Audio\_Visualiser\_DSP.kicad\_sch

Title:

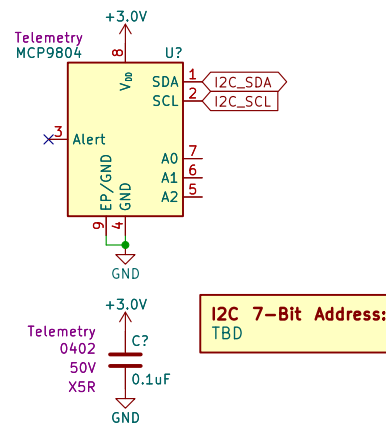
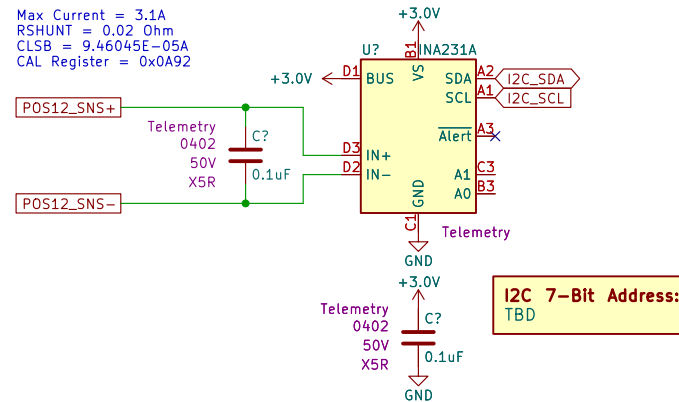
Size: ADate: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:  
Id: 1/18

|                                      |                  |          |
|--------------------------------------|------------------|----------|
| Sheet: /                             |                  |          |
| File: Audio_Visualiser_DSP.kicad_sch |                  |          |
| Title:                               |                  |          |
| Size: A                              | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3                  |                  | Id: 1/18 |





Sheet: /12V Telemetry/  
 File: POS12\_telemetry.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
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 Id: 3/18

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|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /3.3V Power Supply/  
File: POS3P3\_power\_supply.kicad\_sch

Title:

Size: A

Date: 2023-12-25

Rev:

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Id: 4/18

|                                                                   |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
|-------------------------------------------------------------------|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------|--|--|--|--|
| Sheet: /3.3V Power Supply/<br>File: POS3P3_power_supply.kicad_sch |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Title:                                                            |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Size: A                                                           |  |  |  |  | Date: 2023-12-25 |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rev: |  |  |  |  |
| KiCad E.D.A. 10.0.3                                               |  |  |  |  | Id: 4/18         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |

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|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /3.3V Telemetry/  
File: POS3P3\_telemetry.kicad\_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 5/18

|                                                             |                  |          |
|-------------------------------------------------------------|------------------|----------|
| Sheet: /3.3V Telemetry/<br>File: POS3P3_telemetry.kicad_sch |                  |          |
| Title:                                                      |                  |          |
| Size: A                                                     | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3                                         |                  | Id: 5/18 |

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| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /LED Power Supply/  
File: led\_power\_supply.kicad\_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 6/18

|                                                               |                  |          |
|---------------------------------------------------------------|------------------|----------|
| Sheet: /LED Power Supply/<br>File: led_power_supply.kicad_sch |                  |          |
| Title:                                                        |                  |          |
| Size: A                                                       | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3                                           |                  | Id: 6/18 |

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|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /LED Telemetry/  
File: led\_telemetry.kicad\_sch

Title:

Size: A

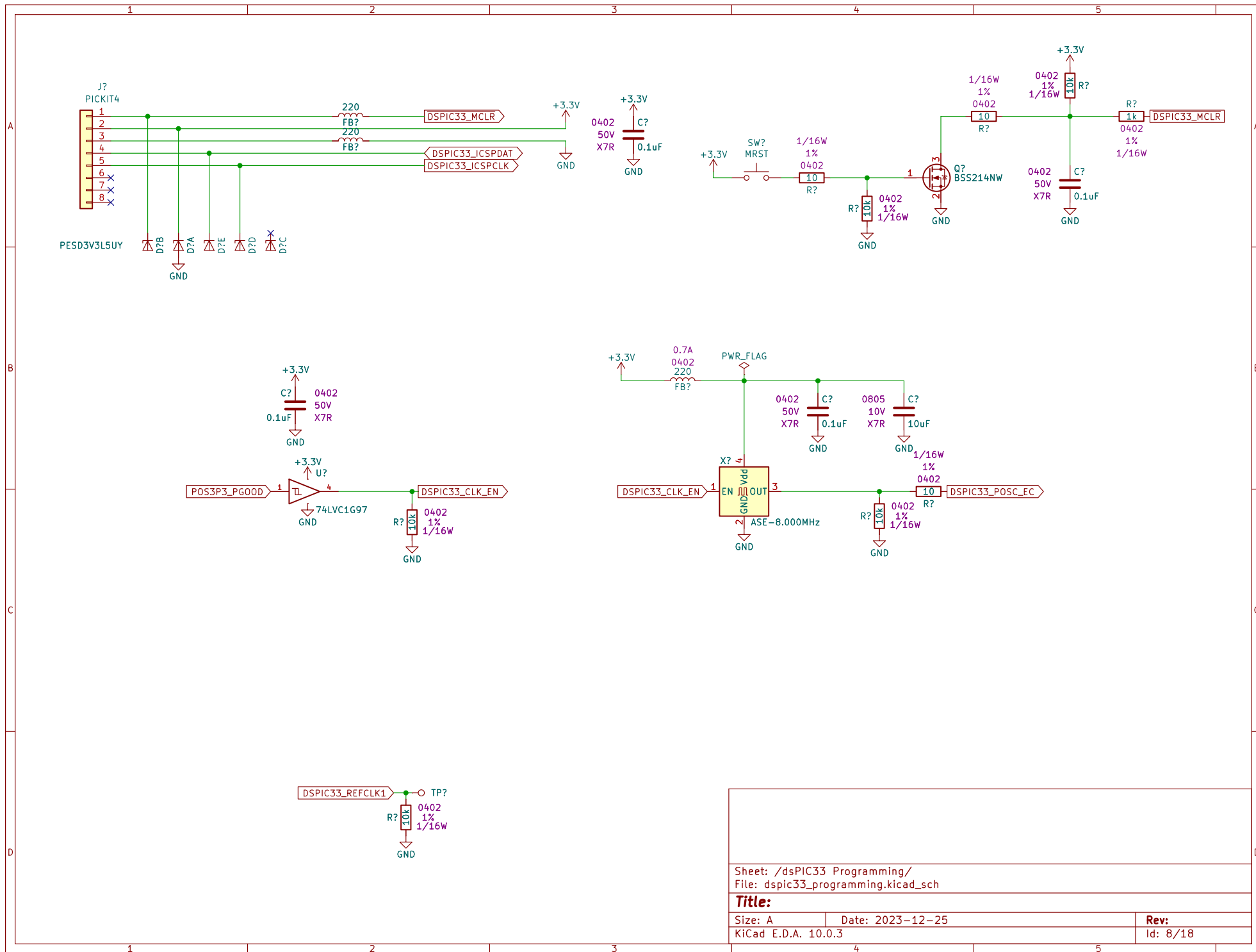
Date: 2023-12-25

Rev:

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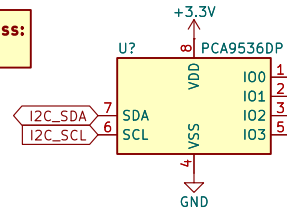
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| File: led_telemetry.kicad_sch |                  |          |
| <b>Title:</b>                 |                  |          |
| Size: A                       | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3           |                  | Id: 7/18 |





# ADD HARDSTRAPS FOR CONFIG STUFF

**I2C 7-Bit Address:**  
0x41



U?B  
dsPIC33AK256MPS103

OA1IN+/AD1AN1

DSPIC33\_AUDIO\_LEFT

OA2IN+/AD2AN1

DSPIC33\_AUDIO\_RIGHT

SDA1/PGD1

SCL1/PGC1

RP23 = REFO1

DSPIC33\_ICSPDAT

DSPIC33\_ICSPCLK

DSPIC33\_REFCLK1

OSCO/CLK0/RP33/IOMAF5/RC0  
OSCI/RP34/IOMAF6/RC1  
PGC3/RP35/PWM4H/IOMAD7/RC2  
RP37/I3CSDA1/IOMAD4/RC4  
PGD3/RP38/I3C\_SCL1/PWM4L/IOMAD6/RC5

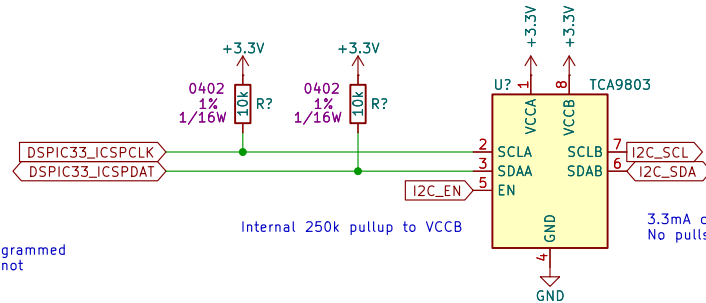
19 USB\_UART\_RX  
20 DSPIC33\_POSC\_EC  
21 USB\_UART\_TX  
22

RP33 = U1RX  
OSCI  
RP35 = U1TX

RP49/PWM2H/IOMAD3/RD0  
TCK/RP50/PWM2L/IOMAD2/RD1  
TD0/RP51/PWM1H/IOMAD1/RD2  
TDI/RP52/PWM1L/IOMAD0/RD3

26 LEFT\_LED\_SPL\_SCK  
27 LEFT\_LED\_SPL\_SDO  
28 RIGHT\_LED\_SPL\_SCK  
29 RIGHT\_LED\_SPL\_SDO

RP49 = SCK1  
RP50 = SDO1  
RP51 = SCK2  
RP52 = SDO2



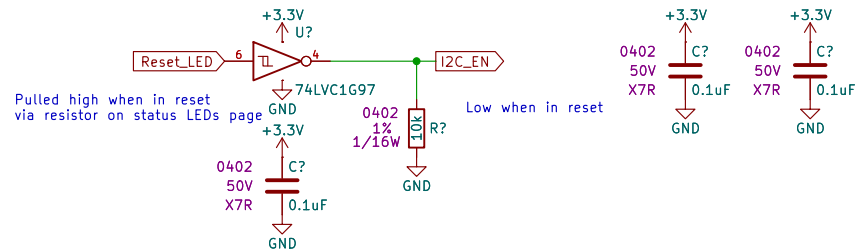
Heartbeat\_LED Reset\_LED

PGOOD\_LED\_SHDN

Error\_LED

CPU\_Trap\_LED

Isolate I2C signals when in reset/being programmed so that ICSP can program the DSC but cannot spook I2C devices with nonsensical data



Sheet: /dsPIC33 GPIO/  
File: dspic33\_gpio.kicad\_sch

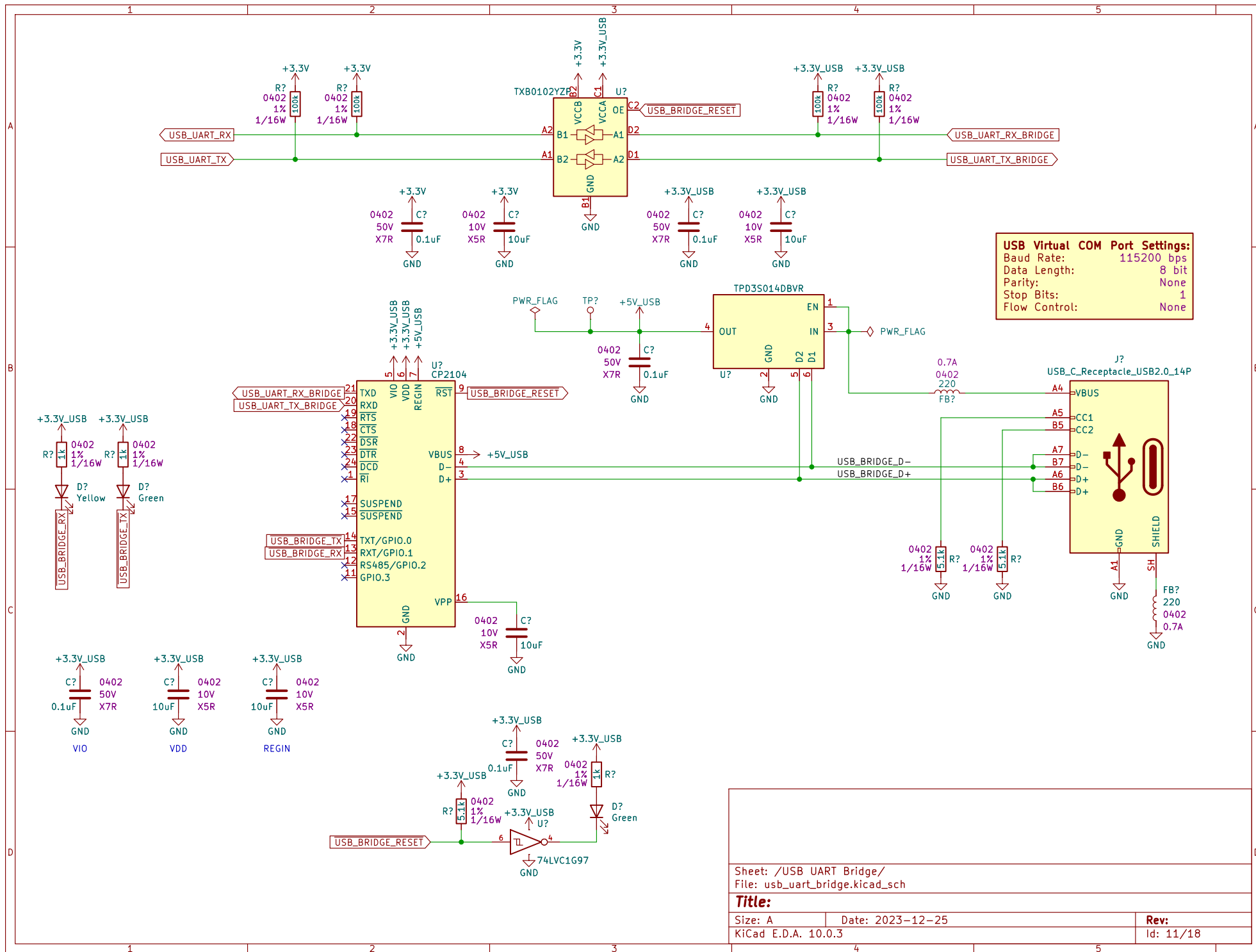
**Title:**

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

**Rev:**

Id: 10/18

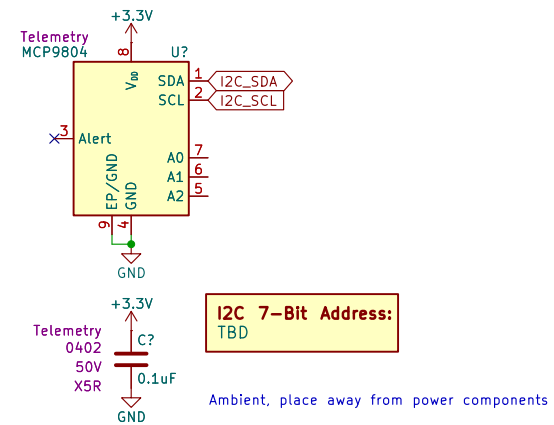
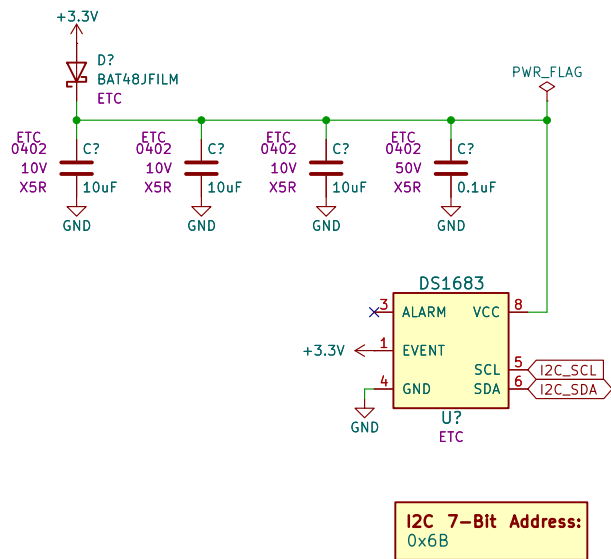


Sheet: /USB UART Bridge/  
File: usb\_uart\_bridge.kicad\_sch

# **Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

Rev:  
Id: 11/18

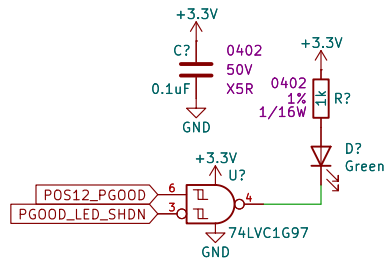
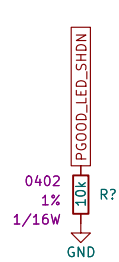


Sheet: /Platform ETC/  
File: platform\_etc.kicad\_sch

# Title:

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

Rev:  
Id: 12/18



Sheet: /PGOOD LEDs/  
File: pgood\_leds.kicad\_sch

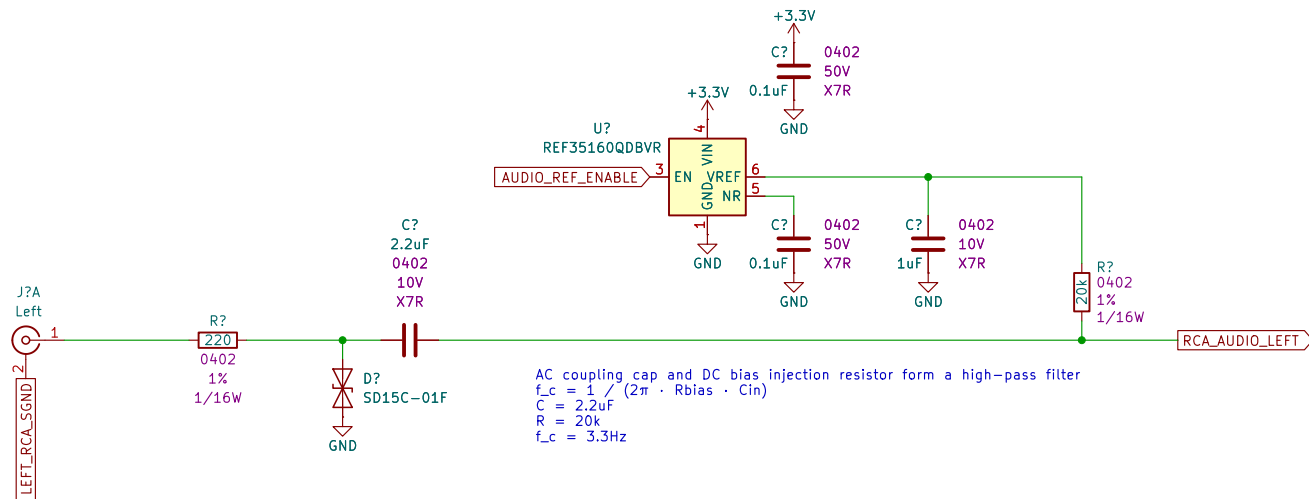
**Title:**

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

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Id: 13/18





Sheet: /RCA Signal Conditioning/  
 File: rca\_signal\_conditioning.kicad\_sch

**Title:**

Size: A Date: 2023-12-25

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 Id: 16/18

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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /Speaker Signal Conditioning/  
File: speaker\_signal\_conditoning.kicad\_sch

Title:

Size: A

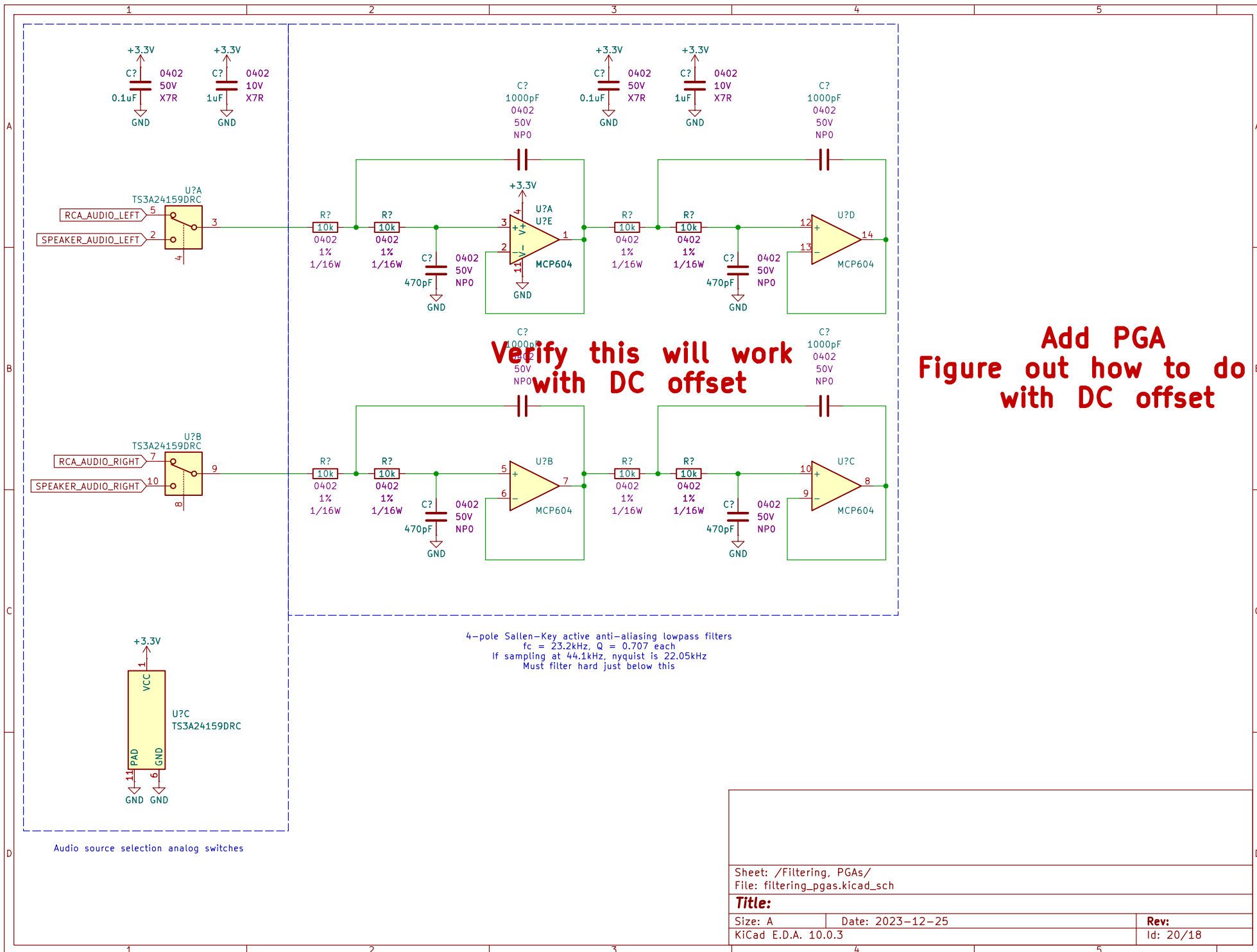
Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 18/18

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|------------------------------------------------------------------------------------|------------------|-----------|
| Sheet: /Speaker Signal Conditioning/<br>File: speaker_signal_conditoning.kicad_sch |                  |           |
| Title:                                                                             |                  |           |
| Size: A                                                                            | Date: 2023-12-25 | Rev:      |
| KiCad E.D.A. 10.0.3                                                                |                  | Id: 18/18 |



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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /Power Button/  
File: power\_button.kicad\_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 21/18

|                                                       |                  |           |
|-------------------------------------------------------|------------------|-----------|
| Sheet: /Power Button/<br>File: power_button.kicad_sch |                  |           |
| Title:                                                |                  |           |
| Size: A                                               | Date: 2023-12-25 | Rev:      |
| KiCad E.D.A. 10.0.3                                   |                  | Id: 21/18 |